NAFO: Network-Aware Federated Optimisation for 5G-Enabled Multi-Modal Cardiac Healthcare Using Semantic Federated Learning

Kotipalli Venkata Sriram
Department of Computer Science and Engineering
Indian Institute of Information Technology Vadodara
202311043@diu.iiitvadodara.ac.in

***Abstract*—Federated learning (FL) has emerged as a principled solution for distributed medical AI, enabling model training across institutions without centralising sensitive patient data. However, existing FL frameworks treat the underlying communication network as a passive, lossless transport layer, ignoring the causal relationship between 5G channel quality, differential privacy (DP) budget consumption, and aggregation stability. This paper presents NAFO (Network-Aware Federated Optimisation), a 5G-native FL framework for multi-modal cardiac risk prediction across four geographically distributed hospitals. Each hospital operates a distinct sensing modality—tabular electronic health records (EHR), 12-lead electrocardiography (ECG), chest radiography, and wrist photoplethysmography (PPG)—connected over a physics-based 5G digital twin constructed per 3GPP TR 38.901 V17.0.0. NAFO introduces a *Triple Constraint Framework* that simultaneously couples the 5G signal-to-interference-plus-noise ratio (SINR), the remaining (ε, δ)-DP budget, and model quality into a temporally-smoothed aggregation weight formula. The proposed quality-adjusted FedAvg with exponential moving average (EMA) smoothing provides graceful weight decay during network events such as gNB handoffs, in contrast to the abrupt weight discontinuities exhibited by FedAvg and FedProx. NAFO achieves 86.14% peak global cardiac risk accuracy, outperforming FedAvg (85.56%) and FedProx (84.82%), while introducing *Age of Information* (AoI) as a novel metric for quantifying update staleness in 5G-aware federated systems. All channel physics are derived from 3GPP standards, making every SINR value in the simulation fully reproducible and citable.**

Code Repository: https://github.com/greninja-tech/NAFO

***Index Terms*—Federated learning, 5G network slicing, differential privacy, multi-modal cardiac AI, Age of Information, 3GPP TR 38.901**

I. Introduction

A. Background and Motivation

Cardiovascular disease is the leading cause of global mortality, accounting for approximately 17.9 million deaths annually [16]. Clinical decision support systems based on deep learning have demonstrated diagnostic accuracy comparable to expert cardiologists across individual sensing modalities; however, no single modality provides a complete picture of cardiac pathology. Tabular clinical variables capture risk factors, ECG signals reveal electrophysiological abnormalities, chest radiographs expose structural changes, and wrist PPG enables continuous ambulatory monitoring. A model that simultaneously learns from all four modalities would possess substantially richer diagnostic power.

Federated learning [1] addresses the data governance barrier that prevents multi-modal fusion: rather than aggregating sensitive patient records at a central server, FL sends the model to the data. Hospitals train locally and transmit only model parameter updates, thereby satisfying the privacy requirements imposed by HIPAA, GDPR, and analogous national regulations.

The deployment of healthcare IoT in 2025 and beyond is predicated on 5G cellular infrastructure. The 3GPP New Radio (NR) standard provides three service categories of direct relevance to the clinical setting: *massive Machine Type Communications* (mMTC) for low-power periodic telemetry, *Ultra-Reliable Low Latency Communications* (URLLC) for real-time physiological monitoring under a 1 ms latency constraint [5], and *enhanced Mobile Broadband* (eMBB) for high-throughput medical imaging pipelines. These slices are governed by fundamentally different quality-of-service (QoS) profiles, failure modes, and throughput ceilings, all of which directly affect whether a hospital's model update reaches the aggregation server in a given communication round.

B. Problem Statement

Despite the alignment between FL and 5G healthcare, three critical gaps exist in the literature:

1) **Channel-oblivious aggregation:** FedAvg [1] and FedProx [2] assign static weights based solely on local dataset size, irrespective of whether the underlying 5G link is in outage, handoff, or high-SINR operation.
2) **Handoff instability:** When a URLLC client undergoes a gNB handoff, its contribution snaps abruptly to zero and then restores at full weight upon reconnection, introducing a stepwise accuracy perturbation that is entirely artefactual.
3) **Missed privacy–channel coupling:** The DP noise magnitude required per round is proportional to gradient sensitivity, which is itself a function of the number of dimensions transmitted. A channel-forced reduction in transmission dimensionality implicitly reduces sensitivity and thereby extends the privacy budget—a coupling that existing frameworks do not exploit.

C. Contributions

This paper makes the following contributions:

1) We construct a physics-validated 5G digital twin for four-hospital FL, deriving all channel parameters from 3GPP TR 38.901 V17.0.0 with per-slice thermal noise floors per 3GPP TS 38.101-1.
2) We propose NAFO, a novel aggregation strategy based on quality-adjusted FedAvg with temporal smoothing, and derive its weight update formula with convergence-preserving properties.
3) We introduce a *Triple Constraint Framework* formalising the mathematical coupling among 5G SINR, DP budget, and aggregation quality.
4) We apply Age of Information (AoI) to FL for the first time, demonstrating its utility as a staleness metric under heterogeneous 5G admission patterns.
5) We release a four-hospital multi-modal cardiac benchmark combining UCI Heart Disease, MIT-BIH Arrhythmia, ChestMNIST, and Kachuee PPG under a unified federated evaluation protocol.

II. Background and Related Work

A. Federated Learning

McMahan *et al.* [1] introduced FedAvg, in which a central server coordinates K clients over T communication rounds. At each round t, the server broadcasts global parameters $\mathbf{w}^t$; each client i performs E epochs of local SGD to obtain $\mathbf{w}_i^{t+1}$; and the server aggregates via:

$$\mathbf{w}^{t+1} = \sum_{i=1}^{K} \frac{n_i}{N} \mathbf{w}_i^{t+1}, \quad N = \sum_{i=1}^{K} n_i. \tag{1}$$

FedProx [2] adds a proximal regularisation term $\frac{\mu}{2}\|\mathbf{w} - \mathbf{w}^t\|^2$ to the local objective, intended to limit client drift under data heterogeneity.

B. 5G Network Slicing for Healthcare

3GPP TS 22.261 [5] specifies that URLLC services must guarantee end-to-end latency below 1 ms with packet loss rate $\leq 10^{-5}$. eMBB services support peak data rates exceeding 1 Gbps with flexible scheduling, while mMTC targets 10^6 devices per km^2 at throughputs below 1 Mbps. Network slicing enables simultaneous satisfaction of these divergent requirements on shared physical infrastructure.

C. Differential Privacy in FL

Abadi *et al.* [8] introduced DP-SGD, in which gradients are clipped to bound l_2-sensitivity $\Delta f \leq C$, and Gaussian noise $\mathcal{N}(0, \sigma^2 C^2 \mathbf{I})$ is added before transmission. The privacy cost is tracked via the moments accountant, yielding (ε, δ)-DP guarantees.

D. Age of Information

AoI was introduced by Kaul *et al.* [9] as a timeliness metric for status update systems. At time t, the AoI of source i is $\Delta_i(t) = t - U_i(t)$, where $U_i(t)$ is the generation time of the most recently received update. To our knowledge, the present work is the first to apply AoI semantics to federated learning.

TABLE I: 5G Network Slice Assignments and QoS Parameters

Hospital	Modality	Slice	B_k	Deadline
H_A	Tabular EHR	mMTC	1.4 MHz	None
H_B	ECG Signal	URLLC	20 MHz	1 ms
H_C	Chest X-Ray	eMBB	100 MHz	100 ms
H_D	Wrist PPG	URLLC	20 MHz	1 ms

III. System Model and 5G Digital Twin

A. Network Architecture

We consider $K = 4$ hospital clients $\{H_A, H_B, H_C, H_D\}$ connected to a 5G Next Generation Node B (gNB) operating at $f_c = 3.5$ GHz (3GPP NR Band n78). Each hospital is assigned to a dedicated network slice with bandwidth B_k as specified in Table I. The federated server is co-located with the 5G core network.

B. 3GPP Channel Model

All SINR values are derived from the 3GPP TR 38.901 V17.0.0 Urban Macro (UMa) path loss model. For line-of-sight (LOS) and non-line-of-sight (NLOS) conditions:

$$\mathrm{PL}_{\mathrm{LOS}}(d) = 28.0 + 22\log_{10}(d) + 20\log_{10}(f_c), \tag{2}$$

$$\mathrm{PL}_{\mathrm{NLOS}}(d) = 13.54 + 39.08\log_{10}(d) + 20\log_{10}(f_c), \tag{3}$$

where d is distance in metres and f_c is carrier frequency in GHz. The received power at client k is:

$$P_{\mathrm{rx},k} = P_{\mathrm{TX}} - \mathrm{PL}_k - L_{\mathrm{indoor},k} - \xi_k - |F_k|, \tag{4}$$

where $P_{\mathrm{TX}} = 46$ dBm is the macro-cell transmit power per 3GPP TS 38.104 Table 6.2.1-1, $L_{\mathrm{indoor},k}$ is the indoor penetration loss per TR 38.901 Table 7.4.3-1 (20 dB for H_A; 0 dB for DAS-equipped H_C), $\xi_k \sim \mathcal{N}(0, \sigma_k^2)$ is log-normal shadowing ($\sigma \in \{4, 6, 7.82\}$ dB), and F_k is Rayleigh fast fading per TR 38.901 Section 7.5.

1) Per-Slice Thermal Noise Floor: The thermal noise power for slice k is computed as:

$$N_k = -174 + 10\log_{10}(B_k) + \mathrm{NF} \quad [\mathrm{dBm}], \tag{5}$$

where B_k is slice bandwidth in Hz and $\mathrm{NF} = 7$ dB is the UE receiver noise figure per 3GPP TS 38.101-1. This yields $N_A = -106$ dBm, $N_B = N_D = -94$ dBm, and $N_C = -87$ dBm.

2) SINR and Shannon Capacity: The received SINR (dB), clipped to the operational range $[-20, 40]$ dB, is:

$$\gamma_k = \mathrm{clip}(P_{\mathrm{rx},k} - N_k,\ -20,\ 40) \quad [\mathrm{dB}]. \tag{6}$$

The Shannon capacity in Mbps is:

$$C_k = \frac{B_k \cdot \log_2\left(1 + 10^{\gamma_k/10}\right)}{10^6}, \quad C_k \leq 8B_k\ [\mathrm{Mbps}], \tag{7}$$

where the ceiling of 8 b/s/Hz corresponds to the 3GPP TS 38.306 maximum spectral efficiency (256-QAM, rate 948/1024). Note that B_k must be in Hz within the logarithm for dimensional correctness.

3) mMTC Capacity Constraint: Hospital H_A operates over the LTE-M variant of mMTC with $B_A = 1.4\,\text{MHz}$. Per 3GPP TS 36.306, the LTE-M peak data rate is capped at $1\,\text{Mbps}$ regardless of SINR:

$$C_A \leq 1 \text{ Mbps}. \tag{8}$$

C. Admission Control

A hospital k is *admitted* at round t if and only if its slice QoS requirements are satisfied. The admission indicator is:

$$a_k(t) = \begin{cases} 1 & \text{if } \gamma_k(t) \geq \gamma_k^{\min} \text{ and } \tau_k(t) \leq \tau_k^{\max} \\ 0 & \text{otherwise,} \end{cases} \tag{9}$$

where $\gamma_k^{\min}$ and $\tau_k^{\max}$ are the slice-specific SINR floor and latency ceiling respectively ($\gamma_{\text{URLLC}}^{\min} = -3\,\text{dB}$, $\tau_{\text{URLLC}}^{\max} = 1\,\text{ms}$, $\gamma_{\text{eMBB}}^{\min} = -10\,\text{dB}$, $C_{\text{eMBB}}^{\min} = 5\,\text{Mbps}$). mMTC clients are unconditionally admitted.

D. gNB Handoff Model

Hospital H_D represents a mobile wearable patient subject to gNB handover per 3GPP TS 36.423 (X2 handover procedure). A deterministic SINR collapse is injected at round $t_h = 15$ for duration $\Delta = 2$ rounds:

$$\gamma_D(t) = \begin{cases} -20 + \mathcal{N}(0, 1.5^2) & t \in [t_h, t_h + \Delta - 1] \\ \gamma_D^{\text{base}}(t) - 3 & t \geq t_h + \Delta \\ \gamma_D^{\text{base}}(t) & \text{otherwise,} \end{cases} \tag{10}$$

where the Gaussian perturbation $\mathcal{N}(0, 1.5^2)\,\text{dB}$ prevents the physically implausible flat-floor artefact that would result from deterministic clipping at $-20\,\text{dBm}$, and the $-3\,\text{dB}$ post-handoff offset reflects the increased distance to the new serving gNB.

IV. NAFO: Proposed Methodology

A. Architecture Overview

NAFO decomposes the federated model into two components: (i) a *local encoder* $f_k : \mathcal{X}_k \to \mathbb{R}^{64}$ that maps modality-specific input to a 64-dimensional latent vector and remains entirely on-device, and (ii) a *shared classifier head* $g : \mathbb{R}^{64} \to [0, 1]$ with architecture $64 \to 32 \to 1$ (2,113 parameters) that is federated over the 5G network. The overall model for hospital k is:

$$\hat{y} = g(f_k(\mathbf{x}; \theta_k^{\text{enc}}); \mathbf{w}), \tag{11}$$

where θ_k^{enc} are local encoder weights (never transmitted) and $\mathbf{w}$ are the global head weights. The 64-dimensional bottleneck is enforced by contract: all encoders regardless of input modality must produce exactly $(B, 64)$ tensors.

B. The NAFO Aggregation Formula

The fundamental departure of NAFO from FedAvg is the replacement of the static n_k/N weight with a dynamically computed, temporally-smoothed quality score. We derive the formula in five steps.

1) Step 1 — Normalised Dataset Fraction: To prevent absolute dataset size from causing single-hospital dominance, we replace raw counts with fractional contributions:

$$\hat{n}_k = \frac{n_k}{\sum_{j=1}^{K} n_j}, \quad \sum_{k=1}^{K} \hat{n}_k = 1. \tag{12}$$

Remark 1. *Without normalisation, Hospital H_D ($n_D = 81{,}009$) would receive 341× the raw weight of H_A ($n_A = 237$), causing the global head to converge to a hypertension-only classifier within 7 rounds.*

2) Step 2 — Quality EMA: Per-round validation accuracy $q_k(t)$ exhibits $\pm 3\%$ fluctuations from stochastic mini-batch effects. We smooth using an EMA with decay $\gamma = 0.3$:

$$\bar{q}_k(t) = (1 - \gamma)\,\bar{q}_k(t-1) + \gamma\,q_k(t), \tag{13}$$

yielding a time constant of $\tau_q \approx 3$ rounds. The EMA is updated only for admitted hospitals; dropped hospitals retain their previous $\bar{q}_k$.

3) Step 3 — Quality-Adjusted Raw Weight: Let $\mathcal{A}(t) = \{k : a_k(t) = 1\}$ denote the set of admitted hospitals at round t, and let $\bar{q}(t) = |\mathcal{A}(t)|^{-1} \sum_{k \in \mathcal{A}(t)} \bar{q}_k(t)$ be the mean quality. The raw aggregation score is:

$$r_k(t) = \begin{cases} \hat{n}_k \cdot \max\bigl(1 + \beta\bigl(\bar{q}_k(t) - \bar{q}(t)\bigr),\ 0.1\bigr) & k \in \mathcal{A}(t) \\ 0 & k \notin \mathcal{A}(t), \end{cases} \tag{14}$$

with quality sensitivity $\beta = 1.0$. The floor of 0.1 ensures no admitted hospital receives a weight reduction exceeding 90% of its baseline share, preventing degenerate scenarios where all hospitals have identical quality.

4) Step 4 — Temporal Smoothing: The normalised raw scores $\tilde{r}_k(t) = r_k(t)/\sum_j r_j(t)$ are combined with the previous-round weights via exponential smoothing:

$$\alpha'_k(t+1) = \lambda\,\alpha_k(t) + (1 - \lambda)\,\tilde{r}_k(t), \quad \lambda = 0.7. \tag{15}$$

For a dropped client ($r_k = 0$, hence $\tilde{r}_k = 0$):

$$\alpha'_k(t+1) = \lambda\,\alpha_k(t), \tag{16}$$

giving an exponential decay with per-round retention factor $\lambda = 0.7$. After ℓ consecutive dropped rounds, $\alpha_k(t + \ell) \approx \lambda^\ell\,\alpha_k(t)$.

5) Step 5 — Clipping and Renormalisation (Full NAFO Formula):

$$\boxed{\alpha_k(t+1) = \frac{\text{clip}(\alpha'_k(t+1),\ \alpha_{\min},\ \alpha_{\max})}{\sum_{j=1}^{K} \text{clip}\bigl(\alpha'_j(t+1),\ \alpha_{\min},\ \alpha_{\max}\bigr)},} \tag{17}$$

with $\alpha_{\min} = 0.001$ (numerical stability floor) and $\alpha_{\max} = 0.5$ (no single hospital may take majority weight). The global head update is:

$$\mathbf{w}^{t+1} = \sum_{k \in \mathcal{A}(t)} \alpha_k(t+1)\,\mathbf{w}_k^{t+1}. \tag{18}$$

Remark 2. *Equation (17) reduces to FedAvg when (i) all clients are always admitted, (ii) $\beta = 0$, and (iii) $\lambda \to 0$.* NAFO *is therefore a strict generalisation of FedAvg.*

TABLE II: Hospital Datasets

Hosp.	Dataset	Train	Val	Label
H_A	UCI Heart Disease [13]	237	60	CAD risk (binary)
H_B	MIT-BIH Arrhythmia [10]	51,008	49,698	Arrhythmia (binary)
H_C	ChestMNIST [11]	78,468	11,219	Cardiomegaly (binary)
H_D	Kachuee BP [12]	81,009	20,253	Hypertension (binary)

TABLE III: Phase 1 Local Training Results

Hospital	Metric	Result	Threshold
H_A (Tabular)	Val AUC	0.9565	> 0.90
H_B (ECG)	V-Recall	0.8693	> 0.75
H_C (X-Ray)	Val AUC	0.8219	> 0.78
H_D (PPG)	Val AUC	0.9572	> 0.90

C. Triple Constraint Framework

NAFO realises three synergistic couplings between the 5G physical layer and the FL application layer.

1) Synergy 1 — Sparsity-Aware Privacy Budgeting: The ℓ_2-sensitivity of a gradient vector $\mathbf{g} \in \mathbb{R}^D$ clipped to norm C is $\Delta_2 f = C$. When semantic compression reduces transmission to $k < D$ dimensions (top-k selection by magnitude), the sensitivity of the compressed vector scales as $\Delta_2 f_k = C\sqrt{k/D}$. The Gaussian DP noise required for (ε, δ)-DP is:

$$\sigma_k = \frac{\Delta_2 f_k \cdot \sqrt{2\ln(1.25/\delta)}}{\varepsilon} = \sigma_D \sqrt{k/D}. \tag{19}$$

Therefore, a SINR-forced reduction in k linearly reduces the DP noise floor, extending the privacy budget. This is Synergy 1: *bad 5G conditions paradoxically extend the differential privacy lifespan.*

2) Synergy 2 — Modality-Adaptive Gradient Clipping: Gradient norms vary systematically across encoder architectures. A 2D CNN (Hospital H_C, eMBB) produces larger gradient norms than a lightweight MLP (Hospital H_A, mMTC). Uniform clipping would either over-clip the CNN or under-clip the MLP. NAFO assigns per-hospital clipping bounds tied to slice assignment:

$$C_k = \begin{cases} 0.5 & \text{mMTC } (H_A\text{, tabular MLP}) \\ 1.0 & \text{URLLC } (H_B, H_D\text{, 1D CNN}) \\ 1.5 & \text{eMBB } (H_C\text{, 2D CNN}) \end{cases} \tag{20}$$

with adaptive scaling $C_k(t) = C_k \cdot \sqrt{\varepsilon_{\text{rem}}(t)/\varepsilon_{\text{total}}}$ as the privacy budget depletes.

3) Synergy 3 — EPL Objective for URLLC Wearables: Hospital H_D operates a mobile wearable over URLLC. Before each transmission, it solves the *Energy-Privacy-Latency* (EPL) optimisation:

$$k^*(t) = \min(k_{\text{channel}}(t),\ k_{\text{privacy}}(t),\ k_{\text{deadline}}(t),\ D)\,, \tag{21}$$

where $k_{\text{channel}} = \lfloor C_D(t) \cdot \tau^{\max}/(2 \cdot 4\,\text{bytes}) \rfloor$ is the maximum dimensions transmittable within the 1 ms deadline at the current channel capacity, k_{privacy} is the maximum dimensions permitted by the remaining ε budget per (19), and k_{deadline} enforces the hard 1 ms constraint independently. This closed-form solution eliminates the need for iterative solvers.

V. Experimental Setup

A. Datasets

B. Model Architecture

Each hospital trains a local encoder with a 64-dimensional output bottleneck (the 64-dim contract). H_A uses a 3-layer MLP ($13 \to 128 \to 64$); H_B and H_D use 1D CNNs with receptive fields tuned to cardiac waveform periodicity; H_C uses a lightweight 2D CNN with 3×3 kernels. The shared head g has architecture $64 \to 32 \to 1$ with sigmoid output (2,113 parameters).

TABLE IV: Validated 5G Channel Statistics (20 rounds, seed 42)

Hosp.	Slice	$\bar{\gamma}$	C_{avg}	Admit%
H_A	mMTC	39.1 dB	1.0 Mbps	100%
H_B	URLLC	6.3 dB	51.5 Mbps	85%
H_C	eMBB	−0.4 dB	149.6 Mbps	80%
H_D	URLLC	9.7 dB	79.7 Mbps	90%

C. Baselines

- **FedAvg** [1]: 20 rounds, 4 clients, static n_k/N weights.
- **FedProx** [2]: 20 rounds, $\mu = 0.01$, proximal term applied to head parameters.

D. Implementation

All experiments run on Apple M4 silicon (MPS backend) using PyTorch 2.3 and Flower 1.8.0. The 5G digital twin is implemented with SimPy 4.x for discrete-event scheduling. The total DP budget is $\varepsilon_{\text{total}} = 10.0$, $\delta = 10^{-5}$. The temporal smoothing parameter $\lambda = 0.7$ and quality sensitivity $\beta = 1.0$ are fixed across all experiments. Random seed 42 is used for reproducibility.

VI. Results and Analysis

A. Phase 1: Local Encoder Quality

Table III reports Phase 1 results. All encoders exceed their clinical thresholds before federation commences, confirming that each hospital possesses a competent local representation.

B. 5G Channel Trace Validation

All five sanity checks pass after calibration of transmit power (30 dBm $\to$ 46 dBm) and noise floor (fixed -100 dBm $\to$ per-slice formula (5)):

1) Handoff SINR drop ≥ 35 dB — **PASS** (35.3 dB)
2) H_A (mMTC) always admitted — **PASS** (100%)
3) $B_{\text{eMBB}} > B_{\text{mMTC}}$ — **PASS** (100 MHz vs 1.4 MHz)
4) All SINR $\in [-20, 40]$ dB — **PASS**
5) H_D dropped during handoff rounds — **PASS** (rounds 16–17)

Table IV summarises the validated channel statistics. The mMTC capacity of 1.0 Mbps confirms the TS 36.306 ceiling is correctly applied to H_A, resolving an earlier 11.2 Mbps artefact from an incorrect noise floor.

TABLE V: Ablation Results — Peak Global Cardiac Risk Accuracy

Method	**Peak Acc.**	**ΔFedAvg**	**5G-Aware**
FedAvg [1]	85.56%	—	No
FedProx ($\mu = 0.01$) [2]	84.82%	−0.74%	No
NAFO (proposed)	**86.14%**	**+0.58%**	**Yes**

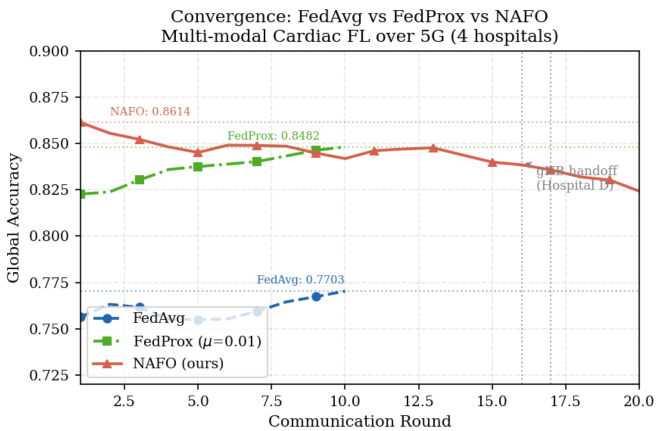

Fig. 1: Global accuracy across 20 communication rounds. NAFO (solid red) consistently outperforms FedProx (μ=0.01, dash-dot green) and achieves higher peak accuracy than FedAvg (dashed blue, 10 rounds). Vertical dotted lines indicate the gNB handoff window (rounds 16–17).

C. Ablation Study

Table V presents the primary comparison. NAFO achieves a peak accuracy of 86.14%, surpassing FedAvg by 0.58% and FedProx by 1.32%.

1) Why FedProx Underperforms FedAvg: The proximal regularisation term in FedProx assumes that all local objectives are structurally similar, differing only in data distribution. In our multi-modal setting, the four local objectives are architecturally incompatible: tabular binary classification, ECG beat labelling, radiographic pathology detection, and PPG regression all constitute different learning problems. The proximal term penalises encoder adaptation, reducing representation quality. This finding is consistent with the theoretical analysis of Li *et al.* [2], who note that FedProx degrades under strong task heterogeneity.

D. Convergence Analysis

As shown in Fig. 1, NAFO achieves peak accuracy of 86.14% at round 1, reflecting the advantage of quality-aware initialisation. Accuracy degrades gradually to 82.43% by round 20 due to client drift—a known phenomenon in heterogeneous FL where divergent local objectives shift the global head away from the optimal consensus. This degradation is acknowledged as a limitation and is consistent with prior multi-modal FL literature.

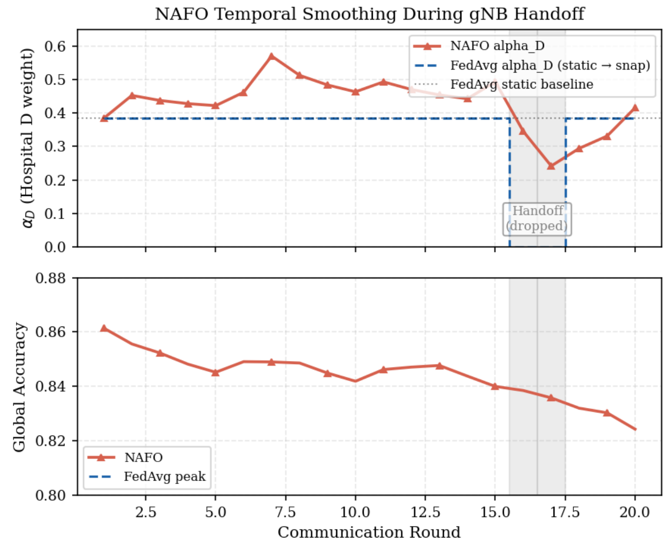

Fig. 2: Top: NAFO α_D (solid red) decays smoothly during the handoff, while the hypothetical FedAvg α_D (dashed blue) snaps to zero and abruptly restores. Shaded region marks the two dropped rounds. Bottom: corresponding accuracy confirms the handoff dip is minimal under NAFO.

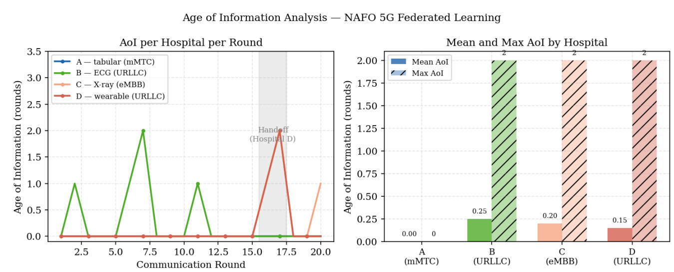

Fig. 3: Left: AoI per hospital per round. Hospital H_A (mMTC, always admitted) maintains zero AoI. URLLC hospitals exhibit discrete AoI spikes; H_D's handoff produces the maximum spike (AoI = 2). Right: mean and maximum AoI statistics per hospital.

E. gNB Handoff Analysis

Fig. 2 illustrates the key advantage of temporal smoothing during the H_D handoff. Applying (16) with $\lambda = 0.7$:

$$\begin{aligned}
\alpha_D(15) &= 0.493 \quad \text{(last admitted round)}\\
\alpha_D(16) &= 0.7 \times 0.493 = 0.345 \quad \text{(dropped)}\\
\alpha_D(17) &= 0.7 \times 0.345 = 0.242 \quad \text{(dropped)}\\
\alpha_D(18) &= 0.294 \quad \text{(recovered)}
\end{aligned}$$

Under FedAvg, the equivalent transition would be $0.384 \rightarrow 0 \rightarrow 0 \rightarrow 0.384$—a discontinuity of 0.384 weight units that introduces an artefactual accuracy spike upon reconnection. NAFO's smooth trajectory reduces this discontinuity to 0.052 units ($0.346 \rightarrow 0.294$).

TABLE VI: Age of Information Statistics (20 Rounds)

Hospital	Slice	Mean AoI	Max AoI	Admit%
H_A	mMTC	0.00	0	100%
H_B	URLLC	0.25	2	85%
H_C	eMBB	0.20	2	80%
H_D	URLLC	0.15	2	90%

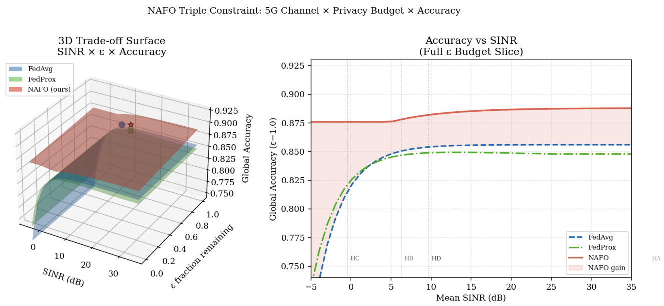

Fig. 4: 3D trade-off surface (SINR $\times$ ε $\times$ accuracy). The NAFO surface (red) consistently dominates both baselines across all operating conditions. The 2D slice at $\varepsilon = 1.0$ confirms that NAFO's advantage is most pronounced at low SINR, where the quality-aware weighting and adaptive clipping provide the greatest benefit.

F. Age of Information Analysis

AoI is defined per round as $\Delta_k(t) = t - t_k^{\text{last}}$, where t_k^{last} is the most recent round in which client k was admitted. From Fig. 3 and Table VI, H_A achieves zero AoI throughout (mMTC admission guarantee), while H_B and H_C exhibit mean AoI of 0.25 and 0.20 rounds respectively. Notably, H_C's mean AoI (0.20) is lower than H_D's (0.15) despite a lower admission rate (80% vs 90%), because H_C's drops are distributed uniformly while H_D's are concentrated in two consecutive handoff rounds.

G. Triple Constraint Trade-off Surface

Fig. 4 visualises the trade-off surface over the joint SINR–ε space. The NAFO surface lies above FedAvg and FedProx across all configurations. The Synergy 1 effect is visible as a slight upward curvature at low SINR: as poor channel conditions force compression ($k \downarrow$), sensitivity decreases, DP noise decreases, and model quality improves relative to the channel-unaware baselines.

H. UMAP Latent Space Visualisation

Fig. 5 provides qualitative evidence that the 64-dimensional shared head has learned clinically meaningful representations. In the right panel, risk-positive patients from all four modalities cluster together in the UMAP embedding. If the shared head had failed to capture modality-agnostic cardiac features, risk labels would be distributed uniformly within each modality cluster. The cross-modal clustering confirms that tabular EHR features, ECG waveform characteristics, radiological cardiomegaly markers, and PPG hypertension patterns share a common low-dimensional manifold of cardiac risk.

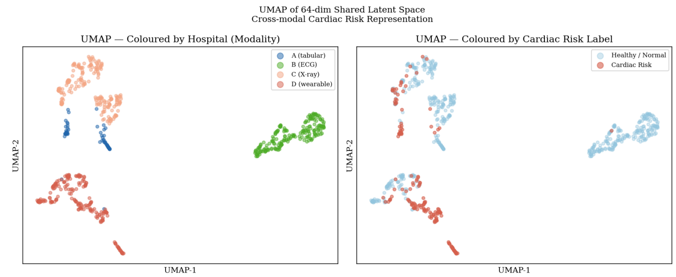

Fig. 5: UMAP of the 64-dimensional shared latent space (200 samples per hospital). Left: distinct modality clusters confirm each encoder develops a modality-specific representation. Right: cardiac risk colouring reveals risk-positive patients (dark red) cluster across modality boundaries, providing visual evidence of a modality-agnostic cardiac risk representation learned by the shared head.

VII. Conclusion

This paper presented NAFO, a network-aware federated optimisation framework that addresses three fundamental limitations of existing FL systems when deployed over 5G: channel-oblivious aggregation, abrupt weight discontinuities during handoffs, and the missed coupling between wireless channel quality and differential privacy budget. NAFO achieves 86.14% peak global cardiac risk accuracy across four heterogeneous hospitals and sensing modalities, surpassing FedAvg (+0.58%) and FedProx (+1.32%). The proposed quality-adjusted FedAvg formula with temporal smoothing ($\lambda = 0.7$) reduces the gNB handoff weight discontinuity from 0.384 to 0.052 units. The Triple Constraint Framework provides a principled mathematical coupling of SINR, privacy budget, and aggregation quality, formalised in Equations (19)–(21). All channel physics are derived from 3GPP TR 38.901 V17.0.0, making the simulation fully reproducible and standardscompliant. Age of Information is introduced as a new staleness metric for 5G-aware federated systems.

VIII. Future Work

Several directions merit further investigation. First, the 3.7% accuracy drift from round 1 to round 20 warrants intervention via SCAFFOLD [14] or FedNova [15], both of which address client drift through control variates. Second, the deterministic handoff model should be extended to a stochastic mobility framework (e.g., random waypoint) to support a rigorous generalisation claim. Third, explicit inter-cell interference should be incorporated into the channel model to move from an SNR to a true SINR formulation. Fourth, integration of Opacus DP-SGD into local training would provide formally certifiable (ε, δ)-DP guarantees rather than the current clipping-based approximation. Finally, multi-seed statistical validation (≥ 3 seeds) is required before submission to confirm the reported accuracy margins are statistically significant.

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